

The P2 lemma: source expansion of $-\det K_2$

A six-scalar factorization, the sign-carrying overlap, and the remainder that is not a restatement

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Abstract

On a real Hermitian invertible matrix A_0 with exactly one negative eigenvalue $-\lambda_- < 0$ and a two-column factor $U = (u, v)$, the Schur matrix $K_2 = I_2 + U^T A_0^{-1} U$ admits the exact factorization

$$\det K_2 = \det(I + R_+) (1 - \gamma/\lambda_-),$$

where R_+ is the Gram of (u, v) in the positive rest form of A_0 and $\gamma = w^T (I + R_+)^{-1} w$ with $w = U^T \psi_-$. The prefactor $\det(I + R_+) \geq 1$ is a sum of positive Gram terms. The sign of $\det K_2$ is carried entirely by the dressed overlap $\gamma - \lambda_-$. This is finite 2×2 algebra, independent of any family. On the nneg-1 branch of the canonical dual CD construction the identity holds live, the r359 fold-pair binding ratio is a *different* 2×2 (already positive at rank $N - 3$), and a raw overlap floor is false (the CD pair may be almost orthogonal to ψ_-). What remains of (P2) is a source lower bound on the relative excess $\gamma/\lambda_- - 1$. Nothing in this document is evidence for or against the Riemann hypothesis.

Status. Lemma 1 is proved (spectral expansion over $\mathbb{R}$). Lemma 2 is proved (matrix-determinant factorization). Lemma 3 is proved (the sign carrier). Lemma 4 is proved (positive structures). Lemma 5 is proved (P1 does not imply P2). Lemma 7 is proved (P2 is not the r359 binding ratio). Proposition 1 is a finite machine check on the nneg-1 branch, not a cofinal law. The original statement $\det K_2 \leq -c_0 < 0$ on the canonical family is *reduced* to the dressed-overlap inequality $\gamma > (1 + \varepsilon)\lambda_-$ of §6. No RH claim. No L^* claim.

1 The object

Let A_0 be real symmetric and invertible, of size $m \geq 2$, with inertia $(m - 1, 1, 0)$: one simple negative eigenvalue $-\lambda_- < 0$ and a positive definite complement. Write ψ_- for a unit eigenvector of $-\lambda_-$, A_+^{-1} for the inverse of A_0 on $\psi_-^\perp$ (zero on $\text{span}\{\psi_-\}$), and $U = (u, v) \in \mathbb{R}^{m \times 2}$. Set

$$K_2 = I_2 + U^T A_0^{-1} U.$$

In the dual CD construction of rounds 367–371 one has $A_0 = R_{N-3} - \frac{1}{2}I$ on the node set Y and U the last two dual Christoffel–Darboux columns. The Haynsworth identity (sorry-free in `RH/Haynsworth.lean`) converts $\text{ind}_-(A_0) = 1$ together with $\det K_2 < 0$ into $A_0 + UU^T \succ 0$. The present note attacks only the determinant sign and floor, by expanding K_2 in source sizes.

2 Six scalars

Write $\alpha = \langle u, \psi_- \rangle$, $\beta = \langle v, \psi_- \rangle$, $w = (\alpha, \beta)^T$, and

$$R_+ = \begin{pmatrix} r_{uu} & r_{uv} \\ r_{uv} & r_{vv} \end{pmatrix} = U^T A_+^{-1} U.$$

The six real scalars are $(\alpha, \beta, \lambda_-, r_{uu}, r_{vv}, r_{uv})$.

Lemma 1 (Spectral expansion).

$$U^T A_0^{-1} U = -\frac{ww^T}{\lambda_-} + R_+.$$

In particular the three quadratic forms of identity (12) of round 371 are

$$\begin{aligned}\langle u, A_0^{-1} u \rangle &= -\alpha^2 / \lambda_- + r_{uu}, \\ \langle v, A_0^{-1} v \rangle &= -\beta^2 / \lambda_- + r_{vv}, \\ \langle u, A_0^{-1} v \rangle &= -\alpha\beta / \lambda_- + r_{uv}.\end{aligned}$$

Proof. The spectral calculus of A_0 on the nneg-1 splitting $\mathbb{R}^m = \text{span}\{\psi_-\} \oplus \psi_-^\perp$ is $A_0^{-1} = -\lambda_-^{-1} \psi_- \psi_-^T + A_+^{-1}$. Sandwiching with U is the claim. On the rational 4-node model $A_0 = \text{diag}(-1, 1, 2, 3)$, $U = [(2, 0, 0, 0)^T, (1, 1, 0, 0)^T]$ one has $(\alpha, \beta, \lambda_-, r_{uu}, r_{vv}, r_{uv}) = (2, 1, 1, 0, 1, 0)$ and the three forms equal $(-4, 0, -2)$ exactly. $\square$

Let $\delta = \alpha v - \beta u$ (the ψ_- -component of the two-form $u \wedge v$) and

$$N = \alpha^2 + \beta^2 + \langle \delta, A_+^{-1} \delta \rangle, \quad \gamma = w^T (I + R_+)^{-1} w.$$

A 2×2 cofactor expansion gives the identity

$$\gamma = \frac{N}{\det(I + R_+)}, \quad N = \alpha^2 + \beta^2 + (\alpha^2 r_{vv} + \beta^2 r_{uu} - 2\alpha\beta r_{uv}).$$

3 Factorization

Lemma 2 (Matrix-determinant factorization). $K_2 = (I + R_+) - ww^T / \lambda_-$ and

$$\det K_2 = \det(I + R_+) (1 - \gamma / \lambda_-). \quad (1)$$

Equivalently, the positive-structure form

$$-\det K_2 = \frac{N - \lambda_- \det(I + R_+)}{\lambda_-} = \det(I + R_+) (\gamma / \lambda_- - 1). \quad (2)$$

Proof. The rank-1 display is Lemma 1 plus $K_2 = I + U^T A_0^{-1} U$. The matrix-determinant lemma on that update is (1). The second display is the same identity rewritten with $N = \gamma \det(I + R_+)$. On the 4-node model, $\det(I + R_+) = 2$, $\gamma = 9/2$, $N = 9$ and $\det K_2 = 2 \cdot (1 - 9/2) = -7$, matching identity (12). $\square$

Lemma 3 (Sign carrier). Assume $\lambda_- > 0$ and $I + R_+ \succ 0$. Then

$$\det K_2 < 0 \iff \gamma > \lambda_- \iff N > \lambda_- \det(I + R_+).$$

The sign of $\det K_2$ is carried by the single scalar $\gamma - \lambda_-$.

Proof. $\det(I + R_+) > 0$ by $I + R_+ \succ 0$, and $\lambda_- > 0$ by nneg-1. The first factor of (1) cannot change sign. $\square$

Lemma 4 (Positive structures). $R_+ \succeq 0$, hence $\det(I + R_+) = 1 + \text{tr } R_+ + \det R_+ \geq 1$. The numerator N is a sum of two nonnegative terms: the raw pair-overlap $\|w\|^2$ and the rest energy $\langle \delta, A_+^{-1} \delta \rangle$ of the wedge component. Neither term is a priori disposable.

Proof. A_+^{-1} is positive semidefinite by construction, so R_+ is a Gram matrix and $I + R_+ \succeq I$. The splitting of N is the 2×2 cofactor identity of §2. Load-bearing of the second summand is Lemma 6 below. $\square$

Identity (12) of round 371 is the expansion of $\det K_2$ in the three quadratic forms; it is compatible with (1) but does not isolate the sign. The wedge term of that identity is the Gram determinant of A_0^{-1} , which mixes the negative mode with the rest; after the spectral split it becomes $\det R_+ - \langle \delta, A_+^{-1} \delta \rangle / \lambda_-$, a difference of positive structures whose sign is stable on the branch (round 371) but is *not* equivalent to $\gamma > \lambda_-$ without the linear terms $1 + \langle u, A_0^{-1} u \rangle + \langle v, A_0^{-1} v \rangle$.

4 The adversarial half

Lemma 5 (P1 does not imply P2). *There exist real symmetric A_0 of inertia $(m-1, 1, 0)$ and columns U with $\det K_2 > 0$. Explicitly, on the 4-node model, replacing U by $0.05 \cdot [(1, 0, 0, 0)^T, (0, 1, 0, 0)^T]$ yields $\gamma = 0.0025 < \lambda_- = 1$ and $\det K_2 = +1.0025 > 0$; the updated matrix $A_0 + UU^T$ keeps a negative eigenvalue.*

Proof. Direct 4×4 arithmetic. Haynsworth is one-sided: both premises are load-bearing (round 367, swapped-premise mutants). $\square$

Algebraically, small overlaps fail to kill ψ_- . A source property of the dual CD columns must forbid $\gamma \leq \lambda_-$ on the nneg-1 branch. A *uniform* raw-overlap floor $\|w\|^2 \geq \kappa_0 > 0$ is the wrong shape: on deep nneg-1 rows the pair-span of ψ_- onto $\text{span}\{u, v\}$ drops to $6 \cdot 10^{-4}$ while $\det K_2$ remains large and negative, because λ_- is smaller still. The comparison that can be uniform is the *relative* excess $\gamma/\lambda_- - 1$.

The sufficient Cauchy bound $\|w\|^2 > \lambda_- (1 + \lambda_{\max}(R_+))$ (which ignores the δ -energy) already fails on a well-conditioned nneg-1 row (kz46: the δ -share of N is 0.723). The wedge dressing of Lemma 4 is load-bearing.

Lemma 6 (The δ -term is necessary). *There is a canonical nneg-1 window at which $\|w\|^2 \leq \lambda_- (1 + \lambda_{\max}(R_+))$ while $\gamma > \lambda_-$. Dropping $\langle \delta, A_+^{-1} \delta \rangle$ from N does not imply (P2).*

Proof. The window kz46 of the sealed r367 ladder: $\lambda_- = 9.869 \cdot 10^{-2}$, $\|w\|^2 = 0.281$, pair-span 0.743, δ -share 0.723, excess $\gamma/\lambda_- - 1 = 0.298 > 0$, and the sufficient Cauchy bound fails. Machine-checked by the companion script. $\square$

5 Not the binding thesis

Round 359 defined $\text{bind} = \lambda_{\min}(S_N) / (\lambda_{\min}(R) - \frac{1}{2})$ for the 2×2 Schur block of $R - \frac{1}{2}I$ at the critical fold pair (2, 4). That ratio consumes the terminal margin by construction.

Lemma 7 (P2 is not bind). *On every resolvable nneg-1 window of the r367 ladder the fold-pair block of A_0 is already positive definite ($\lambda_{\min}$ of the 2×2 pair restriction of A_0 is positive, 45/45). The remaining negative mode ψ_- is rest-hosted on 42/45 (coordinate mass of ψ_- on the fold pair $< 1/2$). Therefore ψ_- is not the r359 pair, and $\gamma > \lambda_-$ does not consume $\lambda_{\min}(S_N)$.*

Proof. The pair/rest split is the r367 negative-direction ledger, reproduced by the companion at w9 ($\text{pmass} = 0.1514$, $\lambda_{\text{pair}} > 0$) and on the branch (45/45 pair blocks positive). The two 2×2 matrices are different objects: last two CD columns against A_0 , versus the fold pair Schur of the full $R - \frac{1}{2}I$. $\square$

A reduction of (P2) to $\text{bind} \geq 1$ would be a restatement of the terminal margin and is forbidden. The shadow test of the excess against $\log |\lambda_{\min}(R_{N-1}) - \frac{1}{2}|$ on the calibration nneg-1 windows is +0.046 (not a rewrite).

6 The reduction

Lemmas 2–4 convert (P2) on the nneg-1 branch into a comparison of two positive scalars. Because $\det(I + R_+) \geq 1$ is SATZ, the inequality

$$\gamma \geq (1 + \varepsilon) \lambda_- \quad (\text{P2}^-)$$

implies $-\det K_2 \geq \varepsilon$. This is strictly stronger than the sign $\det K_2 < 0$, strictly closer to the construction (a Rayleigh quotient of the two-vector $w = U^T \psi_-$ against one rest eigenvalue, no 2×2 Schur determinant of K_2), and does not consume $\lambda_{\min}(S_N)$. It is *not* equivalent to a uniform floor $\|w\|^2 \geq \kappa_0$ (false, §4) and not equivalent to identity (12)’s capture clause (that clause is $\det K_2 < 0$ in other letters, round 371).

The original (P2) with some fixed $c_0 > 0$ is therefore reduced to (P2[−]). A numerical ε is not frozen here: the shallowest well-conditioned nneg-1 window (kz18) has excess 0.2746, so any $\varepsilon \leq 1/4$ is currently compatible with the branch, but a fitted constant is not a source statement.

Proposition 1 (Branch check, not a law). *On the 45 resolvable nneg-1 windows of the r367 ladder, $\gamma > \lambda_-$ holds 45/45, identity (1) holds at f64 grade except two near-singular EXT rows whose λ_- is 10^{-7} (the identity is SATZ; the residual is a spectral reconstruction artefact), $-\det K_2 \in [1.157, 1126.389]$, $\det(I + R_+) \geq 3.307$, excess $\gamma/\lambda_- - 1 \in [0.275, 3.295]$, and the Cauchy sufficient bound holds only 44/45 (Lemma 6). Character-twisted analogs: χ_3 has 21 nneg-1 rows, all sign-stable, w9 already positive-definite (world-separating); χ_4 has 19 nneg-1 rows, all sign-stable. The matched scramble breaks named at (P1) ($\text{ind}_-(A_0) = 21$) while $\det K_2 < 0$ survives. No resolvable canonical window realises Lemma 5.*

The proposition is a machine check, not a cofinal law, and is not the outcome of the round.

7 Named remainder

What (P2[−]) still asks of the source:

The last two dual CD columns, with closed dual weights $u^\vee = c_j(1 - x)/|f|$ (round 356) and Christoffel–Darboux kernel (13) of round 371, have dressed overlap $\gamma = w^T(I + R_+)^{-1}w$ against the unique negative mode ψ_- of $R_{N-3} - \frac{1}{2}I$ strictly larger than λ_- .

The mode ψ_- is an object of the dual ensemble (Borodin complementation, round 356; occupation duality, round 360). It is not node-pinned at the saturated block (round 363) and not hosted by the critical fold pair (Lemma 7). Expanding (α, β) against that mode, via the dual OP restrictions and the kernel K_{N-3} on Y , is the remaining finite identity. No $\lambda_{\min}$ of a Schur block of the full R , and no terminal margin, enters the target.

8 Claim boundary

Finite real algebra on a Hermitian pair (A_0, U) , plus a named remainder on the dual CD construction. Haynsworth remains sorry-free and does not assert (P1) or (P2) on windows. L^{*} and the Riemann hypothesis are not addressed in either direction.